How to Run – OULAD Individual Project (2550723)

What is in this folder

File	Purpose
OULAD_Individual_2550723.ipynb	Main submission notebook – open and run this
submission_outputs.py	Required helper module – must stay in the same folder as the notebook
week4_pipeline.py	Required core ML module – must stay in the same folder as the notebook
requirements.txt	Python package list
studentInfo.csv, studentVle.csv, etc.	Raw OULAD data files

Note on the two .py files: The notebook imports build_submission_outputs from submission_outputs.py, which in turn imports the core pipeline functions from week4_pipeline.py. Both files must be present in the same folder as the notebook or the imports will fail.

Step 1 – Install required packages (do this first)

Open a terminal in this folder and run:

Bash: "pip install catboost pypdf pandas numpy scikit-learn matplotlib seaborn pyarrow ipython"

Why this matters: catboost and pypdf are not part of the standard Anaconda base install. The notebook will fail immediately with ModuleNotFoundError: No module named 'catboost' if this step is skipped.

Step 2 – Open the notebook from this folder

Open Jupyter Notebook or JupyterLab and navigate to this folder. Then open:

OULAD_Individual_2550723.ipynb

Important: The notebook must be opened from the same folder that contains submission_outputs.py, week4_pipeline.py, and the OULAD CSV files. Opening it from a different directory will cause import and data-loading errors.

Step 3 – Run all cells

Use Kernel → Restart & Run All.

The notebook will:

1. Load the six raw OULAD CSV files from the current directory
2. Build the week-4 feature table (~30 seconds on first run)
3. Train Logistic Regression (Pipeline 1) and CatBoost (Pipeline 2)
4. Display all tables (Tables 1–5) and figures (Figs 1–9) as inline cell outputs

All results are visible directly inside the notebook. No separate files need to be opened.

Required raw data files

These six CSV files must be present in the same folder as the notebook:

- studentInfo.csv
- studentRegistration.csv
- studentVle.csv
- vle.csv
- assessments.csv
- studentAssessment.csv